Lesson 1 — Extra Practice

Common Factors: The GCF, Binomial Common Factors and Grouping (Lessons 1 and 2)

Part A — The GCF When a Variable Is Missing

1. State the greatest common factor of each set of monomials. In most sets at least one variable is missing from at least one monomial, and the numerical coefficients have a greatest common factor of their own.

(a) $24a^3b^2$, $36ab^2c$ (b) $45x^4y^3z$, $60x^2z^4$ (c) $42m^5n^2$, $70m^3n^4$, $56m^2$ (d) $18p^2q^3r$, $27pq^4$, $45p^3q^2r^2$ (e) $-48u^3v^4$, $-72u^5v^2w$ (f) $32c^4d$, $-80c^2d^3$, $48cd^2$

2. Two monomials have a greatest common factor of $12x^2y$. One of the two monomials is $36x^3y^2$.

(a) Give two different monomials that could be the second one, and check each.

(b) Show that $48x^2y^3$ cannot be the second monomial.

(c) Explain why a variable that is missing from even one monomial of a set can never appear in that set's GCF.

Part B — Factoring Out a Negative Common Factor

3. Factor each polynomial by removing a **negative** common factor, so that the first term inside the bracket is positive. Every sign inside the bracket changes — check each one by expanding.

(a) $-12x^3 + 18x^2 - 30x$

(b) $-8a^2b - 20ab^2 + 12ab$

(c) $-9m^4 + 15m^3 + 12m^2$

(d) $-24p^3q^2 - 36p^2q^3$

(e) $-x^2y + xy^2 - xy$

(f) $-45u^5v^2 + 27u^3v^4 - 18u^2v^3$

4. The lesson's rule is: *if every monomial is negative, take the GCF negative too.* For each polynomial below, (i) factor removing a **positive** common factor, (ii) factor removing a **negative** common factor, and (iii) say which of the two the rule selects, and why.

(a) $-14y^3 - 21y^2 - 7y$ (b) $-10ab + 15b^2$

(c) One of the two polynomials above has *every* term negative and the other does not. Explain, using your four answers, why that single difference is what decides the sign of the common factor.

Part C — Fractional and Decimal Common Factors

5. Remove a common factor from each polynomial so that the remaining factor is an **integral** polynomial — every coefficient inside the bracket must be a whole number.

(a) $\frac{3}{8}x^2 - \frac{9}{4}x$ (b) $\frac{5}{6}a^3 + \frac{2}{3}a^2 - \frac{1}{2}a$ (c) $\frac{2}{5}m^2n - \frac{4}{15}mn^2$

(d) $0.6x^3 + 0.9x^2 - 0.3x$ (e) $1.2ab^2 + 0.8a^2b$ (f) $-\frac{1}{4}\pi r^3 - \frac{3}{8}\pi r^2 h$

6. Complete the factoring. Check each answer by expanding.

(a) $\frac{4}{9}x^2 - \frac{2}{3}x = _____(2x - 3)$ (b) $0.25t^2 + 0.75t = _____(t + 3)$ (c) $\frac{1}{2}bh + \frac{1}{6}b^2 = _____(3h + b)$

Part D — A Binomial as the Common Factor

7. Factor completely. After you remove the common binomial, check whether the bracket left behind has a common factor of its own.

(a) $3x(y + 2) - 5(y + 2)$ (b) $4a(a - 3) + 12(a - 3)$ (c) $6m(2m + 5) - 9(2m + 5)$

(d) $2x(x-7)+7-x$ (e) $5p(3-q)+10p^2(q-3)$ (f) $(x+1)^2+4(x+1)$

Part E — Factoring by Grouping

8. Factor by grouping. In every part the second pair (once the terms are in a workable order) needs a **negative** monomial removed before the two binomials match.
(a) $12x^2-9x-8x+6$ (b) $10a^2+4ab-15ab-6b^2$ (c) $x^3+5x^2-3x-15$

(d) $6y^3-4y^2-15y+10$ (e) $8m^2-12mn-10mn+15n^2$ (f) $ac-bd+ad-bc$

9. In (a) and (b) the four terms are **out of order**: pairing them as they stand gives two binomials that do not match. In each part, reorder the terms first, then group.
(a) $5x^2+6y-10x-3xy$ (b) $2p^3+15-6p^2-5p$ (c) $12ab-9b+8a^2-6a$

10. Factor completely. Part (a) has a monomial common to all four terms — remove it before you group. In part (c), follow the lesson's practice of leaving the negative inside the bracket rather than rewriting the whole answer.
(a) $3x^3+6x^2+12x+24$ (b) $4a^3-4a^2b-7a+7b$ (c) $-2x^3-8x^2+3x+12$

Part F — Working Backwards

11. Each line below gives a polynomial *already* in factored form. Recover the original polynomial by expanding.
(a) the GCF is $6x^2y$ and the other factor is $2x-3y+4$ (b) $-5ab(3a-2b+1)$ (c) $\frac{1}{5}m(10m^2-5m+2)$

12. In each case the greatest common factor is given. Determine the factor that goes inside the bracket.
(a) $30x^4y^2-45x^3y^3+60x^2y^2$, GCF $15x^2y^2$ (b) A polynomial has GCF $-8m^2n$, and one of its terms is $24m^3n^2$. State the matching term inside the bracket. (c) $21a^3b-14a^2b^2+35a^2b =$ $7a^2b($_____$)$

13. Determine each missing factor or coefficient.

(a) $(m-3)(_____)$ $= 4m^2 + 7m - 12m - 21$

(b) $(_____)(2y+1)$ $= 10y^2 - 6y + 5y - 3$

(c) $6x^2 + 4x + 15x + k$ factors by grouping. Determine k and both factors.

Part G — Find the Error

14. Asked to factor $36x^4y^2 - 24x^3y^3 + 48x^2y^2$ completely, a student wrote

$$36x^4y^2 - 24x^3y^3 + 48x^2y^2 = 6x^2y^2\left(6x^2 - 4xy + 8\right).$$

(a) Expand the student's answer to show it does equal the original polynomial.

(b) The answer is still not finished. State exactly what is wrong with it, and how you can see the problem by looking only at the bracket.

(c) Write the complete factorization, and state the greatest common factor the student should have removed.

15. Three students factor $-12p^3 - 24p^2 - 12p$.

Amira: $-12p\left(p^2 + 2p + 1\right)$, Ben: $-12p\left(p^2 + 2p - 1\right)$, Cass: $-12p\left(p^2 + 2p\right)$.

(a) Expand all three answers and decide which one is correct.

(b) Name the specific mistake Ben made and the specific mistake Cass made. For each, say which of the three original terms was handled wrongly.

(c) Ben says the correct answer is not *complete*. Explain why, at this stage of the unit, it is — and name the tool you would need before you could say anything more about that bracket.

16. Asked to factor $6a^2 - 9a - 4a + 6$ by grouping, a student wrote

$$6a^2 - 9a - 4a + 6 = 3a(2a - 3) + 2(3 - 2a),$$

then stopped, saying "the brackets are different, so this polynomial cannot be factored."

(a) Show that the student's line is correct — expand it and compare with the original.

(b) The conclusion is wrong even though the line is right. Explain the relationship between $(2a-3)$ and $(3-2a)$, and rewrite the second pair so the brackets match.

(c) Finish the factoring.

Part H — Stretch

17. In each of these the common factor is a **power of a binomial**. Take the *lowest* power of the repeated bracket, exactly as you take the lowest exponent of a repeated variable, then simplify what is left.

(a) $(x+3)^3 - 5(x+3)^2$ (b) $6(2y-1)^3 + 9(2y-1)^2$ (c) $8a^2(a-b) - 4a(a-b)^2$

18. A rectangular prism has volume $V = 18x^3y + 24x^2y^2 - 30x^2y$ cubic centimetres, and its height, in centimetres, is the greatest common factor of the three terms.

(a) Write V in factored form and state the height of the prism.

(b) Write an expression for the area of the base of the prism.

(c) Evaluate the height, the base area and the volume when $x = 3$ and $y = 2$, and verify that height $\times$ base area $= V$.

19. *(Multiple Choice)* When $-24a^4b^2 + 36a^3b^3 - 48a^2b^2$ is factored completely, with the greatest common factor removed, so that the first term inside the bracket is positive, the common factor removed is

A. $12a^2b^2$ B. $-12a^2b^2$
C. $-6a^2b^2$ D. $-12a^2b^3$

20. *(Numerical Response)* When $40x^5y^3 - 60x^3y^4 + 100x^3y^3$ is factored completely, the greatest common factor is ax^by^c. The value of $a + b + c$ is _____.

(Record your answer as a three-digit number.)

Lesson 3 — Extra Practice

Factoring Trinomials of the Form $x^2 + bx + c$

Part A — Reading the Signs Before You Hunt

1. Rows 1 and 2 below hide the same two integers, up to sign, and so do rows 3 and 4. Complete the table. The first row is done for you.

Trinomial	**Sign of b**	**Sign of c**	**The two integers**	**Factored form**
$x^2 + 9x + 8$	$+$	$+$	both positive	$(x+1)(x+8)$
$x^2 - 9x + 8$				
$x^2 + 2x - 35$				
$x^2 - 2x - 35$				

(a) Complete the rule: if $c > 0$ the two integers both carry the sign of _____; if $c < 0$ they have _____ signs and the one further from zero carries the sign of _____.

(b) Rows 3 and 4 have the same $|b|$ and the same $|c|$, yet the brackets are different. Explain what changing the sign of b alone does to the pair, and why the *sizes* 5 and 7 never changed.

2. For each trinomial, write the sign pattern of the two integers **first** (from the rule in question 1, before doing any searching). Then name the pair and write the factored form.

(a) $x^2 + 13x + 36$ (b) $x^2 - 13x + 36$ (c) $x^2 + 5x - 36$

(d) $x^2 - 5x - 36$ (e) $x^2 - 35x - 36$ (f) $x^2 + 16x + 39$

3. All four of these are built from the *same* two numbers. Factor each one.

(a) $x^2 + 11x + 28$ (b) $x^2 - 11x + 28$ (c) $x^2 + 3x - 28$ (d) $x^2 - 3x - 28$

(e) In (a) and (b) the middle coefficient is ± 11; in (c) and (d) it is only ± 3. Using the pair 4 and 7, explain where 11 comes from and where 3 comes from.

(f) Is there a trinomial $x^2 + bx + 28$ or $x^2 + bx - 28$ using the pair 4 and 7 that is *not* on this list? Explain why the four above are all of them.

Part B — Take the Common Factor Out First

4. None of these is in the form $x^2 + bx + c$ as written — but every one of them reduces to $x^2 + bx + c$ once the common factor is removed. Factor completely.

(a) $4x^2 + 28x + 40$ (b) $5x^3 - 40x^2 + 75x$ (c) $7x^2 - 7x - 84$

(d) $x^3y - 2x^2y - 35xy$ (e) $6x^4 + 18x^3 - 240x^2$ (f) $2x^2y^2 + 2xy^2 - 112y^2$

5. Each of these has a **negative** leading coefficient. Remove a negative common factor so that what is left starts with $+x^2$, then factor. Watch every sign inside the bracket.

(a) $-x^2 - 9x - 20$ (b) $-3x^2 + 21x - 36$ (c) $-2x^3 + 2x^2 + 60x$

(d) $-x^2 + 4x + 45$ (e) $-5a^2 - 5ab + 150b^2$ (f) $-m^4 + 11m^3 - 24m^2$

6. Two questions about the common-factor step itself.

(a) Verify by expanding that $-x^2 + 4x + 45$ also equals $(9-x)(x+5)$. Say how this form is related to your answer to 5(d), and why leaving the negative inside a bracket is legitimate.

(b) A student tries to start $2x^2 + 7x + 6$ with a common-factor step and gets stuck. Explain why no common factor can turn this one into x^2+bx+c, and what that says about when the common-factor step is available.

Part C — Two Variables

7. Factor completely. The second variable rides along with the second term of each bracket.

(a) $x^2 + 5xy + 6y^2$ (b) $a^2 - 11ab + 18b^2$ (c) $m^2 + 3mn - 54n^2$

(d) $x^2 - 7xy - 60y^2$ (e) $3p^2 - 3pq - 90q^2$ (f) $-4u^2 + 4uv + 80v^2$

(g) A student answers part (a) with $(x+2)(x+3y)$. Expand it and show it is wrong. Then explain, using the y^2 term, why *both* brackets must carry a y.

Part D — Higher Powers in Disguise

8. Each of these is x^2+bx+c wearing a costume: the first term is a perfect square of something, and the middle term is that something to the first power. Factor each one.

(a) x^4+7x^2+12 (b) x^4-10x^2+21 (c) a^4+2a^2-35

(d) x^6+8x^3+15 (e) y^8-y^4-56 (f) $4x^4+20x^2+24$

9. Same idea, but this time at least one bracket factors **again** as a difference of squares. Factor completely — keep going until nothing factors further. Recall $a^2-b^2=(a-b)(a+b)$, while a^2+b^2 does not factor over the integers.

(a) x^4-13x^2+36 (b) x^4-5x^2-36 (c) x^4+3x^2-4 (d) x^4-29x^2+100

(e) Parts (a) and (d) ended with four brackets, but (b) and (c) stopped at three. Name the bracket that stopped the process in (b), and say exactly why it cannot be taken further.

(f) Question 8(a) gave $(x^2+3)(x^2+4)$ and neither bracket went further, yet 4 is a perfect square. Explain why x^2+4 is not a difference of squares.

Part E — When the Method Terminates

10. Factor where possible. Some of these **cannot be factored over the integers** — for those, list every factor pair of c you tested and state that no pair has sum b.

(a) $x^2+7x+13$ (b) $x^2+7x+12$ (c) x^2-6x-8 (d) x^2-6x+8

(e) x^2+4x-6 (f) $x^2+4x-45$ (g) $x^2-9x+18$ (h) $x^2+2x+15$

11. Knowing when to stop.

(a) Explain how you can be certain $x^2 + x + 7$ cannot be factored over the integers after testing only **two** pairs. List them.

(b) Let p be a prime number. Show that $x^2 + bx + p$ can be factored over the integers for exactly two values of b, and state both in terms of p. Test your rule on $p = 13$.

Part F — Working Backwards

12. Consider $x^2 + bx + 24$, where b is an integer.

(a) List every factor pair of 24 (positive and negative), and beside each write its sum.

(b) State every integer value of b for which $x^2 + bx + 24$ can be factored over the integers, and say how many there are.

(c) Every positive value in your list has a negative partner of the same size. Explain why that must happen for *any* positive constant term.

13. One factor is given. Determine the missing coefficient.

(a) $x^2 + 9x + c$ has $(x + 6)$ as a factor. Find c.

(b) $x^2 - 3x + c$ has $(x - 10)$ as a factor. Find c.

(c) $x^2 + bx - 36$ has $(x + 12)$ as a factor. Find b.

(d) $x^2 - 11x + c$ has $(x + 4)$ as a factor. Find c.

(e) $x^2 + kx - 30$ has $(x + 10)$ as a factor. Find k.

(f) $x^2 - x + c$ has $(x + 6)$ as a factor. Find c.

14. *(Numerical Response)* The sum of all **positive** integer values of b for which $x^2 + bx + 18$ can be factored over the integers is _____.

(Record your answer as a three-digit number.)

Part G — Find the Error

15. Asked to factor $x^2 + 11x + 24$, a student wrote $(x + 2)(x + 12)$, reasoning that $2 \times 12 = 24$.

(a) Expand the student's answer to show it is wrong, and name the term that is off.

(b) The student checked the product but not the sum. State the pair that satisfies *both* conditions and write the correct factored form.

(c) 24 has four positive factor pairs. List all four with their sums, and say why checking the product alone can never be enough.

16. Asked to factor $x^2 - 5x + 6$, a second student wrote $(x + 2)(x + 3)$, reasoning that $2 \times 3 = 6$ and $2 + 3 = 5$.

(a) Substitute $x = 1$ into the original trinomial and into the student's answer to show the answer is wrong.

(b) Using the sign table from question 1, state what the signs of b and c here force the two integers to be, and where the student's reasoning broke down.

(c) Factor correctly. Then write the one trinomial that $(x+2)(x+3)$ *is* the factored form of.

Part H — Stretch

17. A **perfect-square trinomial** is one that factors as $(x + n)^2$ or $(x - n)^2$.

(a) Find every value of k for which $x^2 + kx + 36$ is a perfect-square trinomial, and write the factored form for each.

(b) Generalise: for $x^2 + kx + c$ to be a perfect-square trinomial, what must be true of c, and what are the values of k in terms of c?

(c) Is there any integer k making $x^2 + kx + 35$ a perfect-square trinomial? Justify using your rule from (b).

18. Look at this family of factorisations.

$$(x + 2)(x + 3) = x^2 + 5x + 6 \qquad (x + 5)(x + 6) = x^2 + 11x + 30 \qquad (x + 8)(x + 9) = x^2 + 17x + 72$$

(a) In each one the two integers are consecutive. Describe how b and c are built from the smaller integer n.

(b) Use your description to factor $x^2 + 21x + 110$ without hunting through factor pairs. Show the check.

(c) Write the general identity for $(x + n)(x + n + 1)$ in the form $x^2 + bx + c$, with b and c expressed in terms of n.

19. *(Multiple Choice)* Which of the following is a factor of $x^4 - 3x^2 - 40$? A. $x^2 - 5$ B. $x^2 + 5$ C. $x^2 + 8$ D. $x^2 - 4$

Lesson 4 — Extra Practice

Difference of Squares

Part A — Factoring Completely: More Than One Round

1. Each binomial below is a difference of squares whose *first* bracket is itself a difference of squares. Apply the pattern, then look at every bracket and apply it again. Factor completely.

(a) $x^4 - 16$ (b) $a^4 - 1$ (c) $81 - y^4$ (d) $m^4 - 625$

2. These need **three** rounds. Keep going until no bracket is a difference of squares.

(a) $x^8 - 1$ (b) $a^8 - 256$ (c) $256 - k^8$

3. In question 2(a) you factored $x^8 - 1$ into four brackets.

(a) After each of the first two rounds, exactly one bracket could be factored again. Which one — the minus bracket or the plus bracket — and why is it always that one?

(b) State a test you can run on your final answer to be sure you are finished. Then explain how you know $x^4 + 1$ is the end of the line.

Part B — Take the Common Factor Out First

4. None of these is a difference of squares as written — each becomes one only after the greatest common factor is removed. Factor completely.

(a) $2x^2 - 18$ (b) $3x^3 - 27x$ (c) $5a^3 - 80a$

(d) $7p^2q - 63q^3$ (e) $2x^4 - 32$ (f) $a^5 - 16a$

5. Remove the greatest common factor first again. This time **some of them stop there** — once the common factor is out, what is left may be a sum of squares, or may not be a difference of squares at all. Factor completely.

(a) $4x^2 + 36$

(b) $3y^3 + 12y$

(c) $6m^2 - 54$

(d) $2t^4 - 32t^2$

(e) $5x^2 + 45$

(f) $48m - 3m^5$

6. A student writes $2x^2 - 18 = 2(x^2 - 9)$ and stops.

(a) Their line is true, but it is not an answer to "factor completely." Say exactly which rule it breaks, and finish the job.

(b) A second student misses the common factor entirely and writes $2x^2 - 18 = (2x - \ ?\ )(2x + \ ?\ )$. Explain why no pair of integers can fill those blanks.

Part C — Difference or Sum?

7. Factor completely **where possible**. If an expression cannot be factored over the integers, say so and give the reason.

(a) $4m^2 - 25n^2$

(b) $4m^2 + 25n^2$

(c) $8x^2 + 72$

(d) $8x^2 - 72$

(e) $36u^2v^2 + 1$

(f) $36u^2v^2 - 1$

8. *(Error analysis)* A student writes $x^2 + 25 = (x + 5)(x - 5)$.

(a) Expand $(x + 5)(x - 5)$ and state what it actually equals.

(b) Substitute $x = 1$ into both $x^2 + 25$ and the student's answer. What do you get, and what does that prove?

(c) Using "two integers with product 25 and sum 0," explain why *no* factoring of $x^2 + 25$ exists over the integers.

9. *(Error analysis)* Asked to factor $x^4 - 81$ completely, a student writes $x^4 - 81 = (x^2 - 9)(x^2 + 9)$ and stops.

(a) Is the line they wrote correct? Expand it and check.

(b) Finish the question properly.

(c) The two students in questions 8 and 9 made opposite mistakes. Describe each in one sentence.

Part D — Coefficients, Two Variables and Fractions

10. Factor completely. Every term is a perfect square, but the square roots now involve coefficients, two variables, or both.
(a) $16x^4 - 81y^4$ (b) $81a^4 - 16b^4$ (c) $x^4y^4 - 16$ (d) $a^8 - b^8$

11. A fraction can be a perfect square too: $\frac{1}{4} = \left(\frac{1}{2}\right)^2$. Factor each of the following.
(a) $49m^2 - \frac{1}{4}n^2$ (b) $\frac{1}{9}a^2 - \frac{1}{25}b^2$ (c) $\frac{1}{4}p^2 - \frac{9}{16}q^2$

12. Question 11(a) has two correct-looking answers: $\left(7m - \frac{1}{2}n\right)\left(7m + \frac{1}{2}n\right)$ and $\frac{1}{4}(14m - n)(14m + n)$.
(a) Expand both and show they agree.
(b) The second was obtained by removing the fractional common factor $\frac{1}{4}$ first. Which form would you rather hand in, and why? Write 11(c) in that form as well.

Part E — When Each "Square" Is an Expression

13. In $a^2 - b^2 = (a - b)(a + b)$ the letters a and b may stand for whole brackets. Factor each of the following, then **simplify inside each bracket**.
(a) $(x + 3)^2 - 16$ (b) $(y - 5)^2 - 49$ (c) $(2a + 1)^2 - 9b^2$ (d) $(m + n)^2 - (m - n)^2$

14. Same idea, but in most of these, once you simplify the brackets a **common factor appears inside one or both**. Take it out — those are not complete until you do.
(a) $4x^2 - (x + 3)^2$ (b) $9 - (y - 1)^2$ (c) $25p^2 - (p - 2)^2$ (d) $16t^2 - (2t + 6)^2$

15. Factoring is faster than squaring. Evaluate each of the following **without a calculator and without squaring either number** — write it as $a^2 - b^2$, factor, then multiply.
(a) $101^2 - 99^2$ (b) $250^2 - 150^2$ (c) $85^2 - 15^2$ (d) $47^2 - 13^2$

(e) Explain why the same shortcut is useless for $47^2 + 13^2$, and say what that has to do with the fact that a sum of squares does not factor.

Part F — A Stretch, and Two for the Test

16. $x^4 + 4$ is a *sum* of squares, so it is not a difference of squares as it stands — and yet it does factor into two trinomials. The trick is to add and subtract the same term. Work through the steps.

(a) Write $x^4 + 4$ in the form $(\ \ )^2 + (\ \ )^2$ to confirm it really is a sum of squares.

(b) Expand $(x^2 + 2)^2$ and confirm that it equals $x^4 + 4x^2 + 4$.

(c) Since $x^4 + 4 = \big(x^4 + 4x^2 + 4\big) - 4x^2$, rewrite $x^4 + 4$ as $(x^2 + 2)^2 - (2x)^2$ and factor it with the difference of squares.

(d) Check your two trinomials by expanding, and again by substituting $x = 3$ into the original and into your answer.

17. *(Multiple Choice)* When $2x^4 - 32$ is factored completely, one of the factors is
A. $x^4 - 16$ B. $2x^2 - 8$ C. $x^2 + 4$ D. $x^2 - 4$

18. *(Numerical Response)* When $16x^4 - 81y^4$ is factored completely the result is $(ax - by)(ax + by)(cx^2 + dy^2)$, where a, b, c and d are positive integers. The value of $a + b + c + d$ is _____. *(Record your answer in the numerical response box from left to right.)*

Lesson 5 — Extra Practice

Factoring $ax^2 + bx + c$ (where $a \neq 1$)

Throughout this set, factor by **the method of decomposition**: **(1)** compute ac; **(2)** find the pair of integers whose **product is** ac and whose **sum is** b; **(3)** split the middle term with that pair; **(4)** group and remove the common binomial factor. Show all four steps, and **factor completely** — look for a common factor before you decompose.

Part A — Decomposition When ac Is Large

1. Step (2) on its own. For each ac and b below, find the pair of integers with that product and that sum, or write "no such pair". Work through the factor pairs of ac in order — do not guess.

(a) $ac = 120,\ \ b = 22$ (b) $ac = -168,\ \ b = -13$ (c) $ac = 225,\ \ b = -34$

(d) $ac = 72,\ \ b = 11$ (e) $ac = -630,\ \ b = 9$ (f) $ac = 144,\ \ b = -25$

2. Factor using the method of decomposition. Write ac, the pair, the split middle term, and the grouped result on separate lines.

(a) $12x^2 + 25x + 12$ (b) $15x^2 - 34x + 15$ (c) $20x^2 + 31x - 9$

(d) $24x^2 - 29x + 7$ (e) $18m^2 + 9m - 35$ (f) $14y^2 - 53y + 45$

3. Factor. Each one is out of descending order, and two of them have a negative leading term once they are rearranged. *(Follow the booklet and leave the leading negative inside a bracket, as in* $(x+2)(7-6x)$*.)*

(a) $12-31x+20x^2$ (b) $15-2x-24x^2$ (c) $10+11t-6t^2$

4. For $16x^2+34x+15$, $ac=240$.

(a) List every factor pair of 240 made of two *positive* integers, in order, starting with 1×240. How many are there?

(b) Explain why, for this trinomial, you never have to test a pair containing a negative integer.

(c) Factor $16x^2+34x+15$ by decomposition.

Part B — Common Factor First

5. Factor completely. Remove the greatest common factor before you decompose.

(a) $6x^3+21x^2+9x$ (b) $24x^2-50x+24$ (c) $30x^2-4x-16$

(d) $20y^4+2y^3-6y^2$ (e) $4a^3b-22a^2b+30ab$ (f) $60x^2+75x-90$

6. Factor completely. In each of these the leading term is negative, so the common factor you remove is **negative** as well.

(a) $-18x^2+33x-12$ (b) $-8x^2-2x+15$ (c) $-24x^3+2x^2+12x$

7. Question 5(f) was $60x^2 + 75x - 90$.

(a) Compute ac for the trinomial as written, and again for the trinomial left inside the bracket after the common factor comes out.

(b) A candidate pair for step (2) is $\{d,\ ac \div d\}$ for each positive divisor d of $|ac|$. Count the candidates for each of your two values of ac.

(c) In one sentence, say what part (b) shows about *when* to remove a common factor.

Part C — Two Variables

8. Factor. The method is unchanged — the two split pieces each carry one of each variable.

(a) $6x^2 + 11xy - 10y^2$

(b) $12a^2 - 25ab + 12b^2$

(c) $15m^2 + 4mn - 35n^2$

9. Factor completely. Each of these is a common factor away from a two-variable trinomial.

(a) $8p^2q - 2pq^2 - 15q^3$

(b) $24m^2n + 2mn^2 - 15n^3$

(c) In part (a) the two split pieces are $-12pq^2$ and $10pq^2$. State ac for the trinomial inside the bracket and check that pair against it.

Part D — When the Search Comes Up Empty

10. Three of the six trinomials below factor over the integers and three do not. Factor the ones that do. For the ones that do not, list *every* pair you tested with product ac and state that none of them has sum b.

(a) $6x^2 + 13x + 6$

(b) $6x^2 + 5x + 3$

(c) $10x^2 - x - 3$

(d) $9x^2 - 12x + 5$

(e) $4x^2 + 7x - 5$

(f) $15x^2 + x - 6$

11. A classmate says "you can never be sure a trinomial doesn't factor — maybe you just haven't found the right pair yet."

(a) Explain why the search in step (2) always ends after finitely many tests.

(b) For $12x^2 + 7x - 5$, $ac = -60$. Exactly how many candidate pairs are there, and which one works?

(c) Factor $12x^2 + 7x - 5$, then answer your classmate in one sentence.

Part E — Working Backwards

12. Consider $6x^2 + bx + 15$, where b is an integer.

(a) Find **every** value of b for which the trinomial factors over the integers. *(Use the factor pairs of ac — each pair gives one b.)*

(b) Explain why the values of b come in pairs $+b$ and $-b$, and what changes in the factored form when the sign of b flips.

(c) Factor the trinomial for the *smallest positive* value of b you found.

13. Find the missing leading coefficient or missing factor. Check each answer by expanding.

(a) $ax^2 - 11x - 15 = (4x + 3)(_____)$. Find the missing factor and a.

(b) $ax^2 - 17x + 12$ has $(3x - 4)$ as a factor. Find a and the other factor.

(c) $8x^2 + bx - 21$ has $(4x - 7)$ as a factor. Find b and the other factor.

14. $ax^2 + bx + c$ factors as $(3x - 5)(4x + 3)$.

(a) State a, b and c.

(b) State ac and the pair of integers a student would find in step (2).

(c) Write the split middle term, and confirm that grouping it returns the two given factors.

Part F — Find the Error

15. Asked to factor $3x^2 + 4x - 4$, a student wrote

"I need two numbers with product 4 and sum 4, so 2 and 2 : $3x^2 + 4x - 4 = 3x^2 + 2x + 2x - 4$"

and then could not finish.

(a) The student's split is not an arithmetic slip — check that $3x^2+2x+2x-4$ really does equal $3x^2+4x-4$. So which of the four steps went wrong, and what should the product have been?

(b) Group the student's four terms anyway. Write the two brackets you get and say exactly why the method stalls there.

(c) Factor $3x^2+4x-4$ correctly, showing all four steps.

16. Asked to factor $10x^2 - 11x + 3$, a second student wrote

$$10x^2 - 11x + 3 = 10x^2 - 5x - 6x + 3 = 5x(2x - 1) + 3(2x - 1) = (2x - 1)(5x + 3).$$

(a) Expand $(2x - 1)(5x + 3)$ to show the answer is wrong, and say which coefficients it gets wrong.

(b) The split middle term is correct. Name the exact step that is wrong and write the common factor that should have come out of $-6x + 3$.

(c) Finish the factoring correctly, and state the one-sentence rule about the sign of the second group.

Part G — Stretch

17. A trinomial in x^2. Decomposition works exactly as before, with x^2 playing the role of x — and then each bracket factors again.

(a) Factor $4x^4 - 13x^2 + 9$ completely. Decompose first, then examine each bracket. *(Both are a difference of squares — write each as $(a - b)(a + b)$.)*

(b) Factor $9y^4 - 40y^2 + 16$ completely the same way.

(c) Explain how you know the four brackets in part (a) cannot be factored any further, and why a factorisation is not finished at the decomposition stage.

18. Substitution. In $4(m-3)^2+8(m-3)-5$ the repeated bracket $(m-3)$ plays the role of the variable.

(a) Let $u=m-3$ and factor $4u^2+8u-5$ by decomposition.

(b) Replace u with $m-3$ and simplify each bracket.

(c) Check by expanding the original expression into ax^2+bx+c form and factoring it directly.

19. *(Multiple Choice)* One factor of $18x^2+3x-10$ is
A. $3x+2$ B. $3x-2$ C. $6x-5$ D. $2x-3$

20. *(Numerical Response)* The trinomial $12x^2+x-35$ factors as $(ax-5)(cx+7)$, where a and c are positive integers. The value of a^c is _____.
(Record your answer as a three-digit number.)

Lesson 6 — Extra Practice

Further Factoring

A trinomial is a **perfect square trinomial** exactly when **(1)** the *first* term is a perfect square, **(2)** the *last* term is a perfect square (and positive), and **(3)** the *middle* term is **twice the product** of the square roots of the first and last terms (with the matching sign). All three conditions must hold. Throughout this set, **factor completely** — check every bracket, every time.

Part A — Running the Three-Part Test

1. **Do not factor yet.** For each trinomial, state whether condition (1) holds, whether condition (2) holds, and what twice the product of the two square roots would be. Then say whether the trinomial is a perfect square trinomial.

(a) $49x^2 + 28xy + 4y^2$ (b) $16a^2 - 24a + 9$ (c) $25m^2 + 30m + 4$

(d) $36x^4 + 60x^2 + 25$ (e) $9p^2 - 12pq - 4q^2$ (f) $\frac{1}{9}x^2 + \frac{2}{3}x + 1$

2. Each of these fails **exactly one** of the three conditions. Name the condition that fails, then change **one term** so that the trinomial becomes a perfect square trinomial and write its factored form. Three of them can be repaired in two different ways — give both.

(a) $4x^2 + 10x + 25$ (b) $9y^2 - 30y + 100$ (c) $x^2 + 14x - 49$ (d) $49a^2 + 28ab + 16b^2$

3. Read the pattern backwards. Fill in each blank so that the result is a perfect square trinomial, then state the factored form.

(a) $9x^2 + _____ + 49y^2$ (b) $_____ - 20ab + 25b^2$ (c) $16m^4 + _____ + 9$

(d) $\frac{4}{25}x^2 + _____ + 1$ (e) $81 - 90t + _____$ (f) $x^4 + _____ + y^4$

4. Now factor. Every one of these passed the test in question 1's style — confirm the middle term as you go.
(a) $49x^2 + 84xy + 36y^2$ (b) $64a^2 - 112a + 49$ (c) $9m^4 - 30m^2 + 25$

(d) $\frac{1}{4}x^2 + 2xy + 4y^2$ (e) $121p^2 + 66pq + 9q^2$ (f) $100 - 140t + 49t^2$

Part B — Substitution: Give the Block a Name

5. In $a^2 - b^2 = (a - b)(a + b)$ the letter a may stand for a whole expression. **Write "let $A = \ldots$" first**, factor in terms of A, then replace A and simplify inside each bracket.
(a) $(x + y)^2 - 9$ (b) $(2m - n)^2 - 16$ (c) $49 - (a + 3b)^2$ (d) $(3p + q)^2 - 25r^2$

6. Same idea, but now the block sits inside a trinomial. Let A be the repeated bracket, factor the trinomial in A (use **the method of decomposition** where $a \neq 1$), then substitute back and collect like terms inside every bracket.
(a) $2(a + b)^2 + 5(a + b) + 3$ (b) $3(x - 2)^2 - 10(x - 2) + 8$

(c) $(p + q)^2 - 7(p + q) + 12$ (d) $6(m + n)^2 - (m + n) - 15$

7. Here the block is a monomial: let $A = x^2$ (or y^2, or a^2). Factor completely — in every one of these, **both** brackets factor again.
(a) $x^4 - 5x^2 + 4$ (b) $y^4 - 34y^2 + 225$ (c) $x^4 - 17x^2 + 16$ (d) $a^4 - 20a^2 + 64$

8. Three questions about the substitution itself.

(a) A student finishes 6(a) and writes the answer as $(A+1)(2A+3)$. Their factoring is right. Say exactly what is missing, and write the answer properly.

(b) In 6(b) the substitution gives a bracket $A-2$, which becomes $x-2-2$. Explain why you must collect like terms inside every bracket before you stop.

(c) In 7(a) the substitution ends at $(A-1)(A-4)$. Explain why that is not the end of the question, and say how your answer would change if the expression were x^4-5x^2+6 instead.

Part C — One Expression, Several Methods

9. **Common factor, then the disguised trinomial.** Remove the greatest common factor first, then factor what is left — and then look at *those* brackets. Only one of these breaks all the way down into linear brackets; the other two stop earlier.

(a) $2x^4-50x^2+288$ (b) $5a^5-45a^3+100a$ (c) $6x^4-38x^2+40$

10. **Common factor, then the difference of squares — twice.** Write each difference of squares in the order $a^2-b^2=(a-b)(a+b)$.

(a) $3m^5n-243mn$ (b) x^5y-xy^5 (c) $80p^4q-5q^5$

11. **A perfect square trinomial in disguise, and then the bracket again.** Each of these is a perfect square trinomial in x^2, a^2 or m^2. Factor it as a square — then decide whether the bracket inside the square factors further.

(a) x^4-2x^2+1 (b) a^4-18a^2+81 (c) $16x^4+8x^2+1$ (d) $16m^4-24m^2+9$

(e) Two of the four went further and two stopped. State the property of the bracket inside the square that decides it, and use that property to explain each of the four cases in one line.

Part D — Find the Error

12. Asked to factor $x^2 + 10x + 25$, a student wrote

$$x^2 + 10x + 25 = (x + 5)(x - 5),$$

saying "both x^2 and 25 are perfect squares, so it is a difference of squares."

(a) Run the three-part test on $x^2 + 10x + 25$ and state what kind of expression it actually is.

(b) Expand $(x + 5)(x - 5)$, then substitute $x = 1$ into the original and into the student's answer to show the answer is wrong.

(c) Factor correctly. Then write the one sentence about *how many terms* and *which sign* that would have stopped the student.

13. Asked to factor $k^4 - 40k^2 + 144$ completely, a second student let $A = k^2$, factored $A^2 - 40A + 144 = (A - 4)(A - 36)$, substituted back, and wrote

$$k^4 - 40k^2 + 144 = (k^2 - 4)(k^2 - 36).$$

(a) Is the line they wrote true? Expand it and check.

(b) The instruction was "factor completely." Finish the question.

(c) This student and the one in question 12 made opposite mistakes. Describe each in one sentence, then state the check to run on every bracket before writing a final answer.

Part E — Four Terms: Grouping, or Something Else?

14. Every polynomial below has four terms, but they do not all want the same treatment. For each one, say whether you would split it $2 + 2$ (**factoring by grouping**) or $3 + 1$ (three terms forming a perfect square trinomial, leaving a difference of squares). Then factor completely.

(a) $x^2 + 6x + 9 - y^2$

(b) $ax + ay + bx + by$

(c) $3x^3 - 6x^2 + 4x - 8$

(d) $m^2 - n^2 + 10n - 25$

(e) $4a^2 - 12ab + 9b^2 - 16$

(f) $2pq - 6p + 5q - 15$

15. The decision itself.

(a) Describe, in two or three sentences, what you look for *at a glance* to tell a $2+2$ grouping problem from a $3+1$ problem. Mention the number of squared terms and the sign in front of the term that is on its own.

(b) In 14(d) the last three terms are $-n^2+10n-25$. Explain why you must remove a factor of -1 before you can see the perfect square, and write the three terms after you do.

(c) A classmate tries $2+2$ on 14(a): $x(x+6)+(3-y)(3+y)$. Both halves factored, yet the method failed. Say precisely what has to happen after the two halves are factored, and why it does not happen here.

Part F — Stretch

16. **Substitution twice over.** In each expression the repeated block is itself a binomial in x or y. Let A be that block, factor the trinomial in A, substitute back — and then factor each of the two trinomials you get.

(a) $\left(x^2-4x\right)^2-2\left(x^2-4x\right)-15$

(b) $\left(y^2+2y\right)^2-2\left(y^2+2y\right)-3$

(c) Part (b) finishes with a repeated bracket. Name the perfect square trinomial that produced it, and explain why a student who stopped at the A-stage, or at the substitute-back stage, would not have finished — and why stopping at $(y+3)(y-1)(y+1)(y+1)$ would not be wrong.

17. *(Multiple Choice)* When x^4-8x^2+16 is factored completely, the result is

A. $\left(x^2-4\right)^2$ B. $(x-2)^2(x+2)^2$

C. $(x-2)^4$ D. $\left(x^2-4\right)\left(x^2+4\right)$

18. *(Numerical Response)* When $16x^4-72x^2+81$ is factored completely it can be written $(ax-b)^2(ax+b)^2$, where a and b are positive integers. The value of a^b is _____.

(Record your answer as a three-digit number.)

Lesson 7 — Extra Practice

Choosing a Factoring Strategy

The Checklist — Keep This in Front of You

Every question in this set is answered by working through the lesson's checklist, *in order*:

1. **Common factor first.** Always check whether every term shares a common factor. If so, factor it out before doing anything else.
2. **Two terms left (binomial)?** Check whether it is a **difference of squares**, $a^2 - b^2 = (a - b)(a + b)$.
3. **Three terms left of the form $x^2 + bx + c$?** Factor **by inspection** (two integers that multiply to c and add to b).
4. **Three terms left of the form $ax^2 + bx + c$ with $a \neq 1$?** Factor by the **method of decomposition**. Watch for **perfect square trinomials**.
5. **Four terms left?** Try **factoring by grouping**.
6. **Always finish with a check.** Look at each factor and ask: can *this* be factored again?

In this set, step 6 is expanded into a third possible verdict. After the check, every expression is one of: **factored completely**, **not yet finished**, or **does not factor over the integers**.

Part A — Name the Strategy, Do Not Factor

1. **Do not factor these.** For each one, write down (i) the number of terms and (ii) the name of the method you would use — *common factor only*, *difference of squares*, *inspection*, *decomposition*, *grouping*, or *does not factor*.

(a) $9x^2 - 64y^2$ (b) $2ax + 6a - 5x - 15$ (c) $x^2 + 13x + 42$ (d) $15x^2 - 4x - 4$

(e) $12a^3b^2 + 18a^2b^3 - 30a^2b^2$ (f) $p^2 + 49$ (g) $x^4 - 10x^2 + 9$ (h) $49c^2 - 42cd + 9d^2$

2. **Still do not factor.** Each of these needs step 1 before the shape is even visible. Write down (i) the greatest common factor, (ii) how many terms are left inside the bracket, and (iii) which method then applies to what is inside.

(a) $4x^3 - 100x$ (b) $6y^3 + 21y^2 - 45y$ (c) $5m^3 + 5m^2 - 30m$

(d) $2x^3 + 2x^2 - 18x - 18$ (e) $3t^2 + 27$ (f) $-8k^2 + 50$

3. Each expression below is a student's answer — most are *already partly factored*, one was left exactly as it was given. Using step 6 only, state which of the three verdicts applies: **factored completely**, **not yet finished** (say which bracket is the problem and why), or **does not factor over the integers**.

(a) $(x+4)(x^2-9)$ (b) $(2x-7)(3x+1)$ (c) $3(x^2+16)$

(d) $2x(4x^2-12x+9)$ (e) x^2+x+5 (f) $(x-1)(4x+6)$

Part B — Sort First, Factor Later

4. Here are twelve expressions. **Do not factor any of them yet.**

A. x^2-36 **B.** $6x^2+x-12$ **C.** $8x^3y-12x^2y^2$ **D.** $x^2+15x+56$

E. $3ab-6a+5b-10$ **F.** $25m^2-49n^2$ **G.** $x^2-13x+40$ **H.** $10p^2-29p+21$

I. $y^3-2y^2+7y-14$ **J.** $49-t^2$ **K.** $14u^4v^3+21u^3v^4$ **L.** $x^2-2x-48$

Write each letter into exactly one column below. Every column gets at least two letters.

Common factor only	**Difference of squares**	**Trinomial, inspection**	**Trinomial, decomposition**	**Four terms, grouping**

5. Now factor six of them completely: **B**, **E**, **F**, **H**, **I** and **L**. Use the method you assigned in question 4.

(a) $6x^2+x-12$ (b) $3ab-6a+5b-10$ (c) $25m^2-49n^2$

(d) $10p^2-29p+21$ (e) $y^3-2y^2+7y-14$ (f) $x^2-2x-48$

Part C — Two-Step Decisions

6. In each of these the common factor is what *reveals* the difference of squares. Take out the GCF, then keep applying step 2 until no bracket is a difference of squares.
(a) $18x^3 - 32x$ (b) $7a^4b - 7b^5$ (c) $75y - 27y^3$ (d) $50w^3 - 8w$

7. Here the common factor reveals a trinomial with $a \neq 1$. Take out the GCF, then use the method of decomposition on what is left. Show the product ac, the sum b, and your split.
(a) $12x^3 - 2x^2 - 24x$ (b) $30a^2b + 9ab - 21b$ (c) $-18k^3 + 33k^2 - 9k$ (d) $28p^3q - 34p^2q + 6pq$

8. In question 7 the GCF came out first every time. Explain what goes wrong if you try decomposition on $12x^3 - 2x^2 - 24x$ *without* removing $2x$ first — in particular, what value would you even use for a?

Part D — Mixed Set: Decide Every Time

9. Factor completely. The methods are scrambled — run the checklist before you start each one. One of these does not factor at all; say so if you meet it.
(a) $x^2 - 24x + 144$ (b) $4n^2 - 81$ (c) $7ac + 2c - 21a - 6$

(d) $9h^2 + 30h + 25$ (e) $2g^3 - 50g$ (f) $x^2 + 6x + 15$

10. Factor completely. Again the order is scrambled, and again one of them does not factor.

(a) $20r^2 + 7r - 6$ (b) $z^4 - 8z^2 + 15$ (c) $27s^2t - 12t^3$

(d) $ab + 7a + 4b + 28$ (e) $9u^3v^2 - 15u^2v^3 + 21u^2v^2$ (f) $t^2 + 100$

Part E — Knowing When to Stop

11. Each line below is a student's *final* answer. Decide whether it is factored completely. If it is not, finish it.

(a) $2(x^2 - 16)$ (b) $(3x - 1)(3x + 1)$ (c) $5x(x^2 + 2x + 8)$

(d) $(x + 2)(2x^2 - 8)$ (e) $4(x^2 + 9)$ (f) $(x - 3)(x^2 - 5x + 6)$

12. Exactly **four** of the six below do not factor over the integers. Find the two that *do*, factor them, and for each of the other four give the reason (a sum of squares, or no integer pair with the required product and sum).

(a) $x^2 + 5x + 9$ (b) $4x^2 + 25$ (c) $3x^2 + 8x + 2$

(d) $x^2 - x - 1$ (e) $2x^2 - 7x + 6$ (f) $9x^2 - 12x + 4$

Part F — Error Analysis

13. A student is asked to factor $3x^3 + 12x^2 - 2x - 8$. Their work begins: *"$a = 3$, $c = -8$, so $ac = -24$. I need two integers with product -24 and sum 12. . . "*

(a) Which line of the checklist did the student use, and which should they have used instead? Name the two methods.

(b) Factor $3x^3 + 12x^2 - 2x - 8$ completely by the correct method, then check your answer by expanding.

14. A second student writes $16x^2 + 9 = (4x - 3)(4x + 3)$.

(a) Expand the student's answer. What did they actually factor?

(b) Substitute $x = 1$ into both sides to give a one-line disproof.

(c) State the correct verdict for $16x^2 + 9$ and the general rule behind it.

15. A third student factors $8x^2 - 18$ as $(2x - 3)(4x + 6)$.

(a) Expand to show the student's product is genuinely $8x^2 - 18$. So what *is* wrong with the answer?

(b) Give the completely factored form, and state which checklist step this student skipped at the start and which one they failed at the end.

Part G — Stretch

16. Factor $12x^2 + 17x + 6$ by decomposition.

(a) Do it with the middle term split as $8x + 9x$.

(b) Do it again with the split written as $9x + 8x$.

(c) The two groupings look nothing alike. Explain why they must give the same answer.

17. Neither of these is in a helpful order. Rearrange first, then factor completely.

(a) $x^2 + 2y + 2x + xy$ (grouping the first two terms together gets you nowhere — say why, then regroup)

(b) $12x - 9 - 4x^2$ (write it in descending order, take out the common factor including its sign, then look again)

18. *(Multiple Choice)* Which of the following is factored **completely**? A. $2(x^2 - 49)$ B. $(x + 3)(2x - 8)$ C. $3x(x^2 + 5)$ D. $(x - 1)(x^2 - 1)$

19. *(Numerical Response)* When $32x^3 - 50x$ is factored completely it has the form $ax(bx - c)(bx + c)$, where a, b and c are positive integers. The value of $a + b + c$ is _____. *(Record your answer in the numerical response box from left to right.)*

Lesson 8 — Extra Practice

Factoring Polynomial Expressions Unit Review

Part A — Choosing and Applying

1. Work down the Lesson 7 checklist for each expression, but **do not factor yet**. Complete the table. The first row is done for you.

Expression	**Common factor first?**	**Method used on what is left**
$5x^3 - 45x$	yes, $5x$	difference of squares
(a) $12m^2 - 2m - 24$		
(b) $x^3 + 4x^2 - 9x - 36$		
(c) $9a^2 - 30ab + 25b^2$		
(d) $3t^4 - 243$		
(e) $4p^2 + 25q^2$		

2. Now carry out the plans you made in question 1 and factor each expression **completely**. One of the five does not factor — say so, and say why.

(a) $12m^2 - 2m - 24$ (b) $x^3 + 4x^2 - 9x - 36$ (c) $9a^2 - 30ab + 25b^2$

(d) $3t^4 - 243$ (e) $4p^2 + 25q^2$

3. Every expression below opens with $4x^2$ or $4x^4$, and no single method handles all six. Factor each completely.

(a) $4x^2 - 36$ (b) $4x^2 - 36x$ (c) $4x^2 - 36x + 81$

(d) $4x^2 - 36x + 80$ (e) $4x^2 - 36x - 40$ (f) $4x^4 - 36x^2 + 81$

(g) Four of the six start with a common factor and two do not. Explain what you look at **first** to separate (a)–(f), and why the number of terms is a more useful first question than the leading coefficient.

Part B — Factoring Completely

4. Factor completely. Every one takes at least two of the unit's methods — and part (f) keeps its negative *inside* a bracket, as the lesson does.

(a) $5x^3 - 80x$

(b) $12x^4y - 39x^2y - 36y$

(c) $18m^4 - 8m^2n^2$

(d) $-24x^3 + 78x^2 - 18x$

(e) $x^3 - 5x^2 - 49x + 245$

(f) $20 - 11x - 3x^2$

5. Each of these is a trinomial in a **monomial disguise**. Decompose first, then check every bracket for a difference of squares.

(a) $4a^4 - 17a^2 + 4$

(b) $16m^4 - 65m^2n^2 + 4n^4$

(c) $6x^4 + 7x^2 - 20$

(d) $25p^2q^2 - 30pq + 9$

6. Each line below shows a student's *first* step. For each, either finish the factoring or state that the expression is already factored completely — and in every case say which line of the checklist settles it.

(a) $2x^4 - 32 = 2\left(x^4 - 16\right)$

(b) $8x^3 + 40x^2 + 50x = 2x\left(4x^2 + 20x + 25\right)$

(c) $6x^2 - 4x - 16 = 2\left(3x^2 - 2x - 8\right)$

(d) $12y^3 + 8y^2 - 15y - 10 = (3y + 2)\left(4y^2 - 5\right)$

(e) $4x^4 - 37x^2 + 9 = \left(4x^2 - 1\right)\left(x^2 - 9\right)$

(f) $3x^3 + 9x^2 + 2x + 6 = (x + 3)\left(3x^2 + 2\right)$

Part C — Geometry and Applications

7. The area of a rectangle is $A = 6x^2 + 19x + 15$ square centimetres, and both side lengths are binomials with integer coefficients.

(a) Factor A to recover the two side lengths.

(b) For $x > 0$, which of the two is the longer side? Justify your answer without substituting a number.

8. A square metal plate of side $(2x + 3)$ cm has a square hole of side $(x - 1)$ cm cut out of its centre.

(a) Write the remaining area as a difference of two squares, then factor it completely.

(b) Expand both your factored form and the original difference, and check that they agree.

(c) Explain why x must be greater than 1 here.

9. A rectangular prism has volume $V = 4x^3 - 24x^2 + 32x$ cubic centimetres, and each of its three dimensions is one of the factors of V.

(a) Factor V completely and state the three dimensions.

(b) Determine the smallest whole-number value of x for which the prism actually exists, and evaluate V there.

Part D — Reasoning and Error Analysis

10. Decide whether each statement is **true** or **false**. Justify a true statement in one sentence; disprove a false one with a specific counterexample.

(a) Every quadratic trinomial can be factored over the integers.

(b) An expression made of one square *plus* another square can never be factored at all.

(c) $(x-3)(x+3)$ and $(3-x)(3+x)$ are factorings of the same polynomial.

11. Asked to factor $-3x^2+12x+36$ completely, a student wrote $-3x^2+12x+36=-3\left(x^2-4x+36\right)$ and then said the bracket does not factor.

(a) Expand the student's line and show that it is not equal to the original polynomial.

(b) Identify the single term that was handled wrongly, and state the rule about dividing by a negative common factor that it breaks.

(c) Write the complete factoring. Was the student's bracket really unfactorable? Answer for *both* brackets.

Part E — Challenge

12. Each expression below has **four** terms, but grouping them two-and-two leads nowhere. Group them **one and three** instead: three of the terms form a perfect square trinomial, and the whole expression then becomes a difference of squares.

(a) $x^2+12x+36-25y^2$

(b) $m^2-n^2+8n-16$

(c) $9p^2-12pq+4q^2-25$

(d) In part (b) the perfect square is the *second* group and it is being **subtracted**. Explain what happens to the signs of $-n^2+8n-16$ when it is written as $-(n^2-8n+16)$, and why that step is what makes the pattern visible.

13. Working backwards from factors. Determine each unknown constant, then state the complete factoring.

(a) $6x^2 + bx - 15$ has $(3x - 5)$ as one factor. Determine the other factor and the value of b.

(b) $x^3 + ax^2 - 4x - 12$ can be factored by grouping. Determine a, then factor the cubic completely.

(c) $4x^2 + kx + 9$ is a perfect square trinomial. Determine **both** values of k and factor in each case.

14. *(Multiple Choice)* When $x^3 - 4x^2 - 9x + 36$ is factored completely, the result is

A. $(x - 4)\left(x^2 - 9\right)$
B. $(x + 4)(x - 3)(x + 3)$
C. $(x - 4)(x - 3)(x + 3)$
D. $(x - 4)(x - 3)^2$

15. *(Numerical Response)* When $18x^2 - 9x - 14$ is factored completely it can be written $(ax + b)(cx + d)$, where a, b, c and d are integers with $a > c > 0$. The value of $a + b + c + d$ is _____.

(Record your answer as a three-digit number.)